

Luiz Felipe Silva Marques

Email: lfsmarquess@gmail.com
Webpage: luizfelipemath.github.io

EDUCATION

DM–UFPB Universidade Federal da Paraíba

2025–Present

PhD in Mathematics

João Pessoa, Brazil

- Research interests: Elliptic Partial Differential Equations, Geometric Analysis, Stability of Geometric and Functional Inequalities.
- Advisor: Prof. João Marcos Bezerra do Ó

IME–USP Universidade de São Paulo

2020–2024

BSc in Mathematics with Honorable Mention for Outstanding Academic Performance

São Paulo, Brazil

- Average grade: 9.6/10.
- Undergraduate thesis: Several Complex Variables and Hartog’s Theorem.
- Advisor: Prof. Paulo Domingos Cordaro.

PUBLICATIONS

1. Simulating Simple Random Walks with a Deck of Cards

R. Alves, S. Estácio, S. Frómeta, M. Jara, R. Marinho, **L. F. S. Marques**, and J. V. A. Pimenta.

Brazilian Journal of Probability and Statistics, 39(2):272–286, 2025.

RESEARCH EXPERIENCE

Undergraduate Research Project in Analysis

2022–2024

IME–USP Universidade de São Paulo

São Paulo, Brazil

- Supervised by Prof. Paulo Domingos Cordaro.
- Supported by the FAPESP Scientific Initiation Scholarship.
- Studied integration theory, L^p spaces, topological vector spaces, distribution theory, Fourier analysis, and Sobolev spaces.

Mathematics Research Days for Undergraduates

Jan–Mar 2024

ICMC–USP Universidade de São Paulo

São Paulo, Brazil

- Supported by a FAPESP scholarship.
- Participated in a summer research program for undergraduate students.
- Collaborated on a research project in probability theory that resulted in the publication “Simulating Simple Random Walks with a Deck of Cards” (*Brazilian Journal of Probability and Statistics*, 2025).

Exchange semester in Germany

Oct 2022–Mar 2023

TUM Technische Universität München

Munich, Germany

- Supported by the USP Merit Scholarship.
- Participated in a six-month academic exchange program.
- Completed graduate-level coursework in Functional Analysis and Probability Theory, demonstrating adaptability to the German academic environment.
- Gained academic experience in an international research environment.

Undergraduate Research Project in Dynamical Systems

2020–2022

IME–USP Universidade de São Paulo

São Paulo, Brazil

- Supervised by Prof. Sônia Regina Leite Garcia and Prof. Manuel Valentim de Pera Garcia.
- Supported by the USP Scientific Initiation Scholarship.
- Studied hyperbolicity, symbolic dynamics, topological conjugacy, structural stability, chaos, bifurcation theory, and related topics in discrete dynamical systems.

SCHOLARSHIPS AND AWARDS

CAPES PhD Scholarship

2025–Present

Funding for PhD studies in Mathematics

Honorable Mention for Outstanding Academic Achievement

2024

Awarded upon graduation from the BSc program in Mathematics at USP

ICMC–USP Summer Program Scholarship

2024

Mathematics Research Days for Undergraduates

FAPESP Scientific Initiation Scholarship

2022–2023

Undergraduate Research Project in Analysis

Grant [22/01477-4](#)

USP Merit Scholarship

2023

Grant for six-month exchange at TUM, Germany

USP Scientific Initiation Scholarship

2020–2022

Undergraduate Research Project in Dynamical Systems

IMECC–UNICAMP Summer Program Scholarship

2022

Linear Algebra course (final grade: A+)

Bronze Medal – Elon Lages Lima Math Competition

2021

Organized by the Brazilian Mathematical Society (SBM)

TEACHING ASSISTANTSHIPS

Linear Algebra I at DM–UFPB

2026.1

Undergraduate course

Linear Algebra II at DM–UFPB

2025.2

Undergraduate course

Calculus I at IME–USP

2024.1

Undergraduate course

Analysis in $\mathbb{R}^n$ at IME–USP

2023.2

Undergraduate course

General Topology at IME–USP

2023.1

Graduate course

Introduction to Number Theory at IME–USP

2022.1

Undergraduate course

Calculus II for Physics at IF–USP

2021.2

Undergraduate course

TALKS

Locally Convex Topological Vector Spaces	2023
31st SIICUSP International Symposium on Scientific and Technological Research, USP	
The Krein-Milman Theorem and Some Applications	2023
Functional Analysis student seminar group, IME-USP	
Recurrence and Transience in Simple Random Walks	2023
S4 student seminar group, IME-USP	
Symbolic Dynamics: an Introduction Through the Logistic Function	2022
S4 student seminar group, IME-USP	

ACADEMIC SERVICE

1st Pure Mathematics Week at IME-USP	2024
Co-founder and Organizer	
Organized the first edition of an undergraduate-led academic event promoting pure mathematics within IME-USP.	
Undergraduate Mathematics Program Committee at IME-USP	2022
Elected Student Representative	
Represented undergraduate students in committee discussions concerning curriculum and program development.	

EDUCATIONAL OUTREACH

Volunteer Mathematics Tutoring	2021–2023
Mathematics Tutor at Salvaguarda	
Provided free mathematics tutoring to Brazilian public high school students through a nationwide educational initiative.	

LANGUAGES

- English:** C1
- Portuguese:** Native